PhD Thesis Colloquium

Intelligent Reflecting Surfaces for Wireless Communications: Low-Complexity Techniques and Performance Analysis

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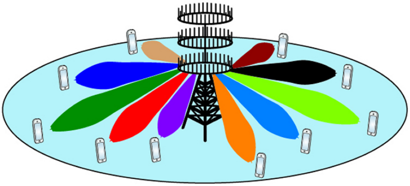
Setting the Stage:
Intelligent Reflecting Surface (IRS)

Array Signal Processing for Wireless Communications

- Multiple antenna systems: A breakthrough in the evolution of wireless communications
  - 5 - 10x boost in spectral efficiency

- Antenna arrays exploit spatial DoF!
  1. **Beamforming**
  2. **Spatial diversity**
  3. **Spatial multiplexing**



A system with multiple antenna BS†

- Array signal processing for wireless comm.: adapt to the wireless channel!

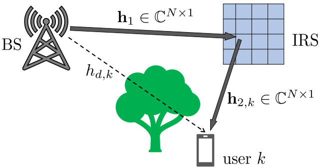
Key Question

Can we **control** the wireless channel instead of just adapting?

† Emil Björnson *et al.*, *Massive MIMO Networks: Spectral, Energy, and Hardware Efficiency*, Foundations and Trends in Signal Processing, 2017.

IRS: A Game Changer for 5G and Beyond †

- Intelligent reflecting surface:
  - Meta-surface, an array of passive elements
  - Reflects signals in required directions
  - **Controls wireless channels** as per the need
- A.k.a. reconfigurable intelligent surface (RIS)
- Advantages: SNR, coverage, energy efficiency
- A hot research topic:
  - Approx. 10K articles on IEEE Xplore
  - ETSI study items, QComm prototypes, etc.



IRS-aided wireless system

Channel:

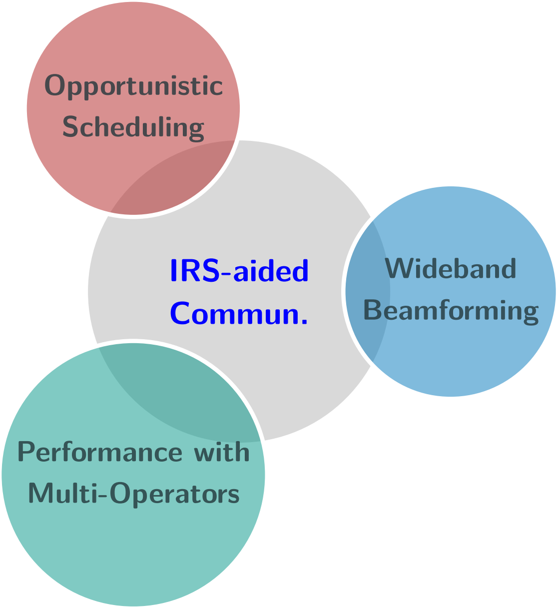
$$\boxed{h_k = \mathbf{h}_{2,k}^T \boldsymbol{\Theta} \mathbf{h}_1 + h_{d,k},}$$

IRS configuration:

$$\boxed{\boldsymbol{\Theta} = \mathrm{diag}\left(e^{j\phi_1}, \ldots, e^{j\phi_N}\right)}$$

† E. Basar, M. Di Renzo, J. De Rosny, M. Debbah, M.-S. Alouini, and R. Zhang, "*Wireless communications through reconfigurable intelligent surfaces*," IEEE Access, vol. 7, pp. 116753–116773, 2019.

This Thesis: Low-complexity Schemes & Performance Analysis



PART I: IRS-aided Opportunistic Communications

Optimizing the IRS and Beamforming SE

Beamforming SE: Performance of Optimized IRS

The **beamforming SE** at UE-k with optimized IRS is

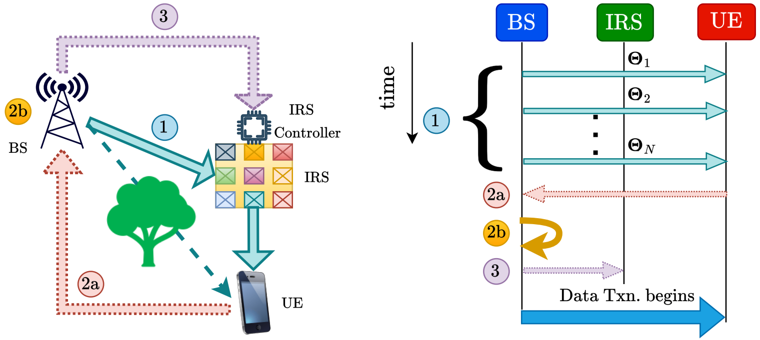
$$R_k^{BF} = \log_2\left(1 + \frac{P}{\sigma^2}\left|\sqrt{\beta_{r,k}}\sum_{n=1}^{N}|h_{1,n}||h_{2,k,n}| + \sqrt{\beta_{d,k}}|h_{d,k}|\right|^2\right).$$

It is achieved when

$$\theta_{n,k}^{BF} = \angle h_{d,k} - \angle(h_{1,n} \times h_{2,k,n}), \qquad n = 1, \ldots, N.$$

- **Array/BF gain** of N-element IRS scales as $\mathcal{O}(N^2)$:
  - IRS enables **constructive superposition** of signal copies at UE

Three-Fold Overheads & Research Question



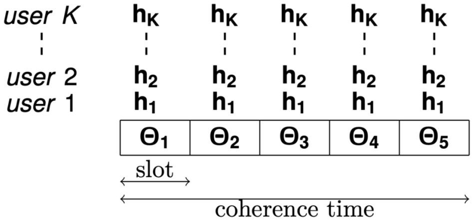
1 Channel estimation
2b Phase optimization
2a CSI feedback
3 Phase transportation

Problem for PART - 1

How can we obtain **optimal benefits without optimizing the IRS** and avoiding the three-fold overheads?

Solution: Opportunistic Communications

Idea: Randomly tune the IRS configurations and schedule one of K UEs opportunistically having the best channel in every time slot!



$$k^* = \arg\max \frac{R_k(t)}{T_k(t)}$$

Avg. rate

$$R_k(t) = \log_2\left(1 + \frac{P|h_{k,q}(t)|^2}{\sigma^2}\right)$$

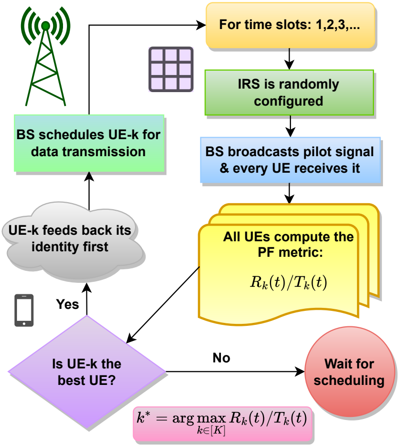
Proportional-fair scheduling

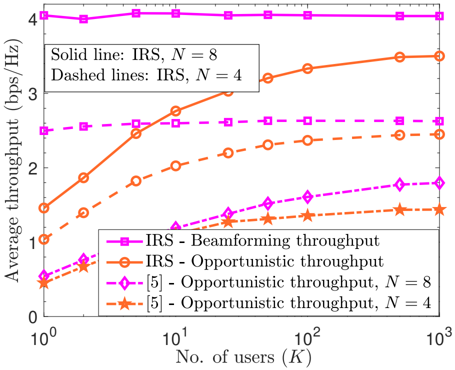
Key Result: Exploit multi-user diversity

With a large # UEs, a random IRS config. is optimal to at least 1 UE, opportunistic scheduling of UEs yields BF-SE:

$$\lim_{K\to\infty}\left(R_K^{opp} - \frac{1}{K}\sum_{k=1}^{K} R_k^{BF}\right) = 0$$

Random IRS becomes Near-Optimal IRS - The Overall Scheme





- **# UEs** depends on **IRS phase distbn.**, **channel stat.**

<u>**Feedback scheme**</u>: V. Shah, N. B. Mehta, R. Yim, "**Optimal Timer Based Selection Schemes**" in *IEEE Transactions on Communications*, June 2010.

[5] P. Viswanath, D. N. C. Tse and R. Laroia, "**Opportunistic beamforming using dumb antennas**" in *IEEE Transactions on Information Theory*, June 2002.

Spatially I.I.D. vs. LoS Channels

Features	**Spatially i.i.d. Channels**	**LoS Channels**
Channel distbn.	i.i.d., $\mathcal{CN}(\cdot)$ - Rayleigh	Array steering vector
Sampling distbn.	$\mathcal{U}[0, 2\pi)$	$\theta_i = \pi(i-1)\sin(\phi)$, $\phi \sim \mathcal{U}[0, 2\pi)$
# UEs required	$K \geq \log\left(\frac{1}{1-P_{\mathrm{succ}}}\right)\left(\frac{\pi}{\epsilon}\right)^{N}$	$K \geq \log\left(\frac{1}{1-P_{\mathrm{succ}}}\right)\left(\frac{\pi}{\epsilon}\right)$

- OC requires fewer UEs with LoS due to smaller DoR

Opportunistic Comm. with Spatial Correlation at IRS

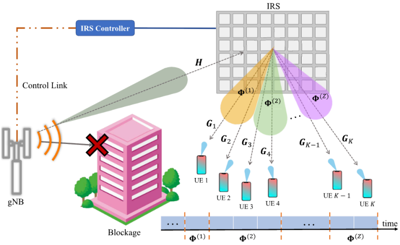
Can we **generalize** the results to arbitrary **spatial correlation**?

- Spatial correlation: local scattering/ mutual coupling, clustering of UEs
- Low-dimensional decomposition:

$$\mathbf{h}_{2,k} = \boldsymbol{\Sigma}_k^{1/2}\tilde{\mathbf{h}}_{2,k} \approx \boldsymbol{\Sigma}^{1/2}\tilde{\mathbf{h}}_{2,k},$$

with $\tilde{\mathbf{h}}_{2,k} \sim \mathcal{CN}(0, \mathbf{I}_M),\ M = \mathrm{rank}(\boldsymbol{\Sigma})$

- Channel BEM: $\mathbf{h}_{2,k} = \sum_{m=1}^{M} \alpha_{k,m}\mathbf{u}_{k,m}$



Clustered UEs

Key Questions

1. What is an appropriate sampling distribution at the IRS?
2. How many UEs are sufficient?

How to Determine the Sampling Distribution?

Functional optimization problem w.r.t. density function $f_{\boldsymbol{\theta}}(\boldsymbol{\theta}')$

$$\underset{f_{\boldsymbol{\theta}}(\boldsymbol{\theta}')}{\arg\max} \quad \bar{R} \triangleq \mathbb{E}_{\boldsymbol{\theta},\mathbf{h}_{f,k}}\left[\log_2\left(1+|\boldsymbol{\theta}'^H\mathbf{h}_{f,k}|^2 P/\sigma^2\right)\right], \quad \text{(P0)}$$

$$\text{s.t.} \int_{\boldsymbol{\theta}'\in\mathbb{U}^N\bigcup\{1\}} f_{\boldsymbol{\theta}}(\boldsymbol{\theta}')\mathrm{d}\boldsymbol{\theta}' = 1, \quad \text{(C1)}$$

$$\text{and} \int_{[\boldsymbol{\theta}']_2}\ldots\int_{[\boldsymbol{\theta}']_{N+1}} f_{\boldsymbol{\theta}}(\boldsymbol{\theta}')\mathrm{d}\boldsymbol{\theta}' = \delta([\boldsymbol{\theta}']_1 - 1) \quad \text{(C2)}$$

- The channel process across UEs is jointly stationary & ergodic
- (C1): PDF normalization; (C2): uncontrollable direct link
- Phase mapping: $\mathcal{F}: \mathbf{h}_{f,k} \longmapsto \left[1, \left(\exp\left\{j\left(\angle\mathbf{h}_{r,k} - \angle h_{d,k}\right)\right\}\right)^T\right]^T$

The Optimal Sampling Distribution

Optimal Density Function

$$f_{\boldsymbol{\theta}}^{\mathrm{opt}}(\boldsymbol{\theta}') = \int_{\mathcal{F}^{-1}(\boldsymbol{\theta}')} \delta\left(\mathbf{h} - \mathcal{F}^{-1}(\boldsymbol{\theta}')\right) p_{\mathbf{h}_f}(\mathbf{h}) \mathrm{d}\mathbf{h},$$

where $\mathcal{F}^{-1}(\boldsymbol{\theta}') \triangleq \{\mathbf{h} \in \mathbb{C}^{N+1} : \mathcal{F}(\mathbf{h}) = \boldsymbol{\theta}'\}$, and $p_{\mathbf{h}_f}(\mathbf{h})$ is the PDF of the cascaded channel, $\mathbf{h}_f = \left[h_d, (\mathbf{h}_1 \odot \mathbf{h}_2)^T\right]^T$.

- Sampling distbn. matches the channel distbn. under $\mathcal{F}(\cdot)$ map
- **Proof outline**:
  1. Law of iterated expectations to formulate a conditional functional
  2. Upper & lower bound the functional via $\ell_1 - \ell_\infty$ Hölder inequality; apply the Sandwich Theorem; obtain the optimal conditional PDF
  3. Marginalize the optimal conditional PDF w.r.t. the channel distbn.

What is the Success Rate of the Scheme?

- Success at UE-k: $\mathcal{E}_k^\delta \triangleq \left\{ |\boldsymbol{\theta}^H \mathbf{h}_{f,k}|^2 \geq (1-\delta) \|\mathbf{h}_{f,k}\|_1^2 \right\}, \delta \in (0,1)$
- **Overall success**: $P_{\mathrm{succ}} = \Pr\left(\bigcup_{k=1}^{K} \mathcal{E}_k^\delta \right) = 1 - \prod_{k=1}^{K} \left(1 - \Pr\left(\mathcal{E}_k^\delta \right) \right)$

Success Probability of the Scheme using a PF Scheduler

$$P_{\mathrm{succ}} \geq 1 - \left(1 - \prod_{m=1}^{M} \frac{1}{1 + \frac{1-\delta}{\delta} \cdot \frac{\lambda_m}{\lambda_M}} \right)^K \sim 1 - \mathcal{O}\left(\left(1 - \delta^M \right)^K \right)$$

where $\lambda_1 \geq \ldots \geq \lambda_M$ are the non-zero eigenvalues of $\boldsymbol{\Sigma}$.

Proof steps: a) Low-dimensional projection of random vectors

b) Prob. of reverse triangle inequality: $\Pr\left(|\sum_i X_i| \geq (1-\delta) \sum_i |X_i| \right)$

c) Tail prob. of Rayleigh quotient of a heteroscedastic Gaussian vec.

How many UEs are Sufficient?

Number of UEs for a Target Success Rate

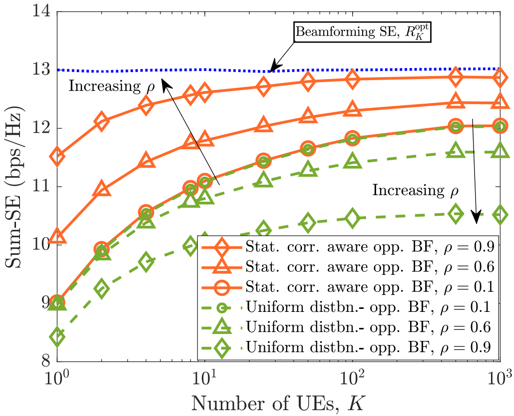
$$\text{If } K \geq K^* \triangleq -\log\left(1 - P_{\mathrm{succ}}\right) \prod_{m=1}^{M} 1 + \left[\frac{(1-\delta)}{\delta} \cdot \frac{\lambda_m}{\lambda_M}\right]$$

$$\sim \mathcal{O}\left(-\left[\log\left(1 - P_{\mathrm{succ}}\right)\right]\left(\frac{1}{\delta}\right)^{M}\right),$$

then, we have "$(1-\delta)N^2$ success" in every time slot.

- Number of UEs scales with **rank of the covariance matrix**
- The degrees of randomness reduce with correlation
- For a fixed dimensional space, # UEs does not depend on N

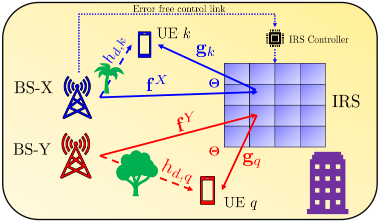
Numerical Studies: We do not need so many UEs!



- Spatial correlation positively helps the OC scheme
  - E.g., 20 UEs are sufficient to obtain 95% of the BF gain!
- Sampling Distribution @ IRS is **important**!

PART - II: IRS-Aided Multiple Operator Systems

Problem: Passive IRS Reflects Signals of All Operators



IRS-aided two-operator system

- **Two operators**: X & Y, e.g., Airtel & Jio: Txn. in 2 different frequencies
- Op. X deploys and controls an IRS
- IRS **does not have band-pass filter**
- IRS reflects all signals:
  - Optimal IRS @ in-band UEs
  - Random IRS @ out-of-band UEs

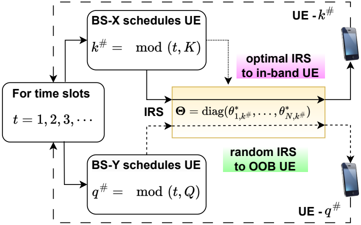
Problem for PART - 2

How does an IRS deployed by a mobile operator **affect the performance** of an **out-of-band (OOB) operator**?

Performance in Sub-6 GHz Bands (FR-1: 410 MHz - 7.125 GHz)

An IRS Does not Degrade the OOB Performance!

- Op. X serves K UEs, Op. Y serves Q UEs using TDMA
- Round-robin (RR) scheduling of UEs
- Op. X configures the IRS for a scheduled UE in every time slot
- **Qns.** Ergodic SE? Outage prob.?



RR scheduling

Ergodic sum-SE of the operators

$$\textbf{MO-X:}\ \bar{R}^{(X)} \approx \frac{1}{K}\sum_{k=1}^{K}\log_2\left(1+\left[N^2\beta_{r,k}c_1 + Nc_2+\beta_{d,k}\right]\frac{P}{\sigma^2}\right)$$

$$\textbf{MO-Y:}\ \bar{R}^{(Y)} \approx \frac{1}{Q}\sum_{q=1}^{Q}\log_2\left(1+\left[N\beta_{r,q} + \beta_{d,q}\right]\frac{P}{\sigma^2}\right)$$

IRS Benefits the OOB Operators Instantaneously Also

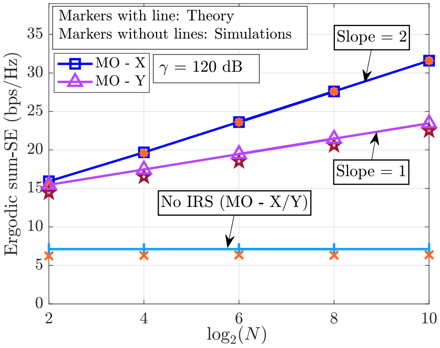
- OOB instant. gain: $Z_N^{(Y)} \triangleq \left| \sum_{n=1}^{N} f_n^Y g_{q,n} + h_{d,q} \right|^2 - |h_{d,q}|^2 \mathbb{1}_{\{N \neq 0\}}$

Complementary Cumulative Distribution Function of $Z_N^{(Y)}$

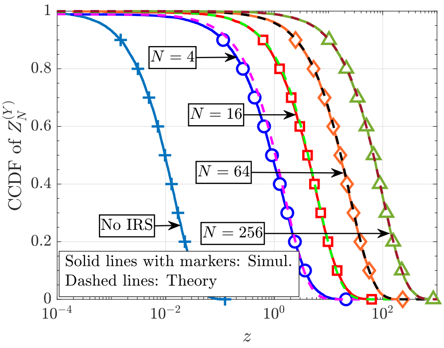
$$\bar{F}_{Z_N^{(Y)}}(z) = \begin{cases} 1 - \dfrac{1}{N\tilde{\beta}+2} \times e^{\left(\frac{z}{\beta_{d,q}}\right)}, & \text{if } z < 0, \\ \left(\dfrac{N\tilde{\beta}+1}{N\tilde{\beta}+2}\right) \times e^{-\left(\frac{z}{\beta_{d,q}\left(1+N\tilde{\beta}\right)}\right)}, & \text{if } z \geq 0. \end{cases}$$

- $\bar{F}_{Z_N^{(Y)}}(0) \longrightarrow 1$: The IRS positive impacts the OOB UEs, a.s.
- OOB channel gain in the **presence** of an IRS **stochastically dominates** the OOB channel in **the absence** of an IRS

Numerical Studies: How does the IRS help an OOB channel?



SE vs. IRS elements



Instantaneous channel gain

- IRS facilitates reception of additional signal copies @ OOB UE
- IRS enhances the rich scattering in the OOB channel
- **IRS does not degrade the OOB perf. - It only helps!**

Performance in mmWave Bands (FR-2: 24.25 GHz - 52.6 GHz)

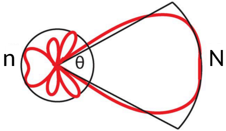
Modelling the Channels at mmWave Frequencies

- **Directional** due to high attenuation & non-rich scattering
- **Saleh-Valenzuela Model**:

$$\mathbf{f}^p = \sqrt{\frac{N}{L_{1,p}}} \sum_{i=1}^{L_{1,p}} \gamma_{i,p}^{(1)} \mathbf{a}_N^*(\phi_{i,p}); \;\; \mathbf{g}_\ell = \sqrt{\frac{N}{L_{2,\ell}}} \sum_{j=1}^{L_{2,\ell}} \gamma_{j,\ell}^{(2)} \mathbf{a}_N^*(\psi_{j,\ell})$$

- Array steering vector: $\mathbf{a}_N(\omega) = \frac{1}{\sqrt{N}} \left[1, e^{-j\pi\omega}, \ldots, e^{-j\pi(N-1)\omega}\right]^T$
- **Beam resolution** capability of an array:
  - N element ULA forms N resolvable beams
  - Resolvable beam book and angle book:

$$\mathcal{A} \triangleq \{\mathbf{a}_N(\phi), \phi \in \mathbf{\Phi}\} \, ; \mathbf{\Phi} \triangleq \left\{ \left(-1 + \frac{2i}{N} \right) \right\}_{i=0}^{N-1}$$



  - Uniform distribution over beam book:

$$\mathcal{U}_{\mathcal{A}}(\phi) = \frac{1}{|\mathbf{\Phi}|} \mathbb{1}_{\{\phi \in \mathbf{\Phi}\}} = \frac{1}{N} \mathbb{1}_{\{\phi \in \mathbf{\Phi}\}}$$

Case 1: LoS Scenario

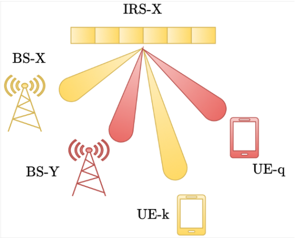
- LoS channel at the in-band UE:

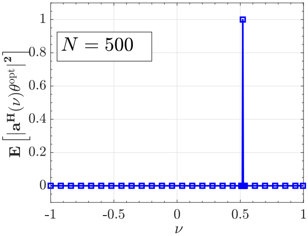
$$h_k = N\gamma_{X,k}\dot{\mathbf{a}}_N^H(\omega_{X,k}^1)\boldsymbol{\theta} + h_{d,k}$$

- Optimal IRS configuration:

$$\boldsymbol{\theta}^{\mathrm{opt}} = \frac{h_{d,k}\gamma_{X,k}^*}{|h_{d,k}\gamma_{X,k}|} \times N\dot{\mathbf{a}}_N(\omega_{X,k}^1)$$

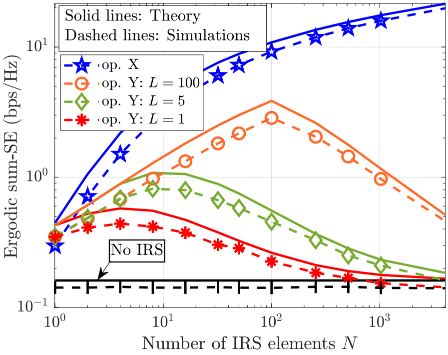
- IRS focuses in the direction of in-band UE with N^2 array gain
- L cascaded paths: With probability $\frac{L}{N}$, the IRS aligns to OOB UE
- With probability $1 - \frac{L}{N}$, the IRS does not alter the OOB channel



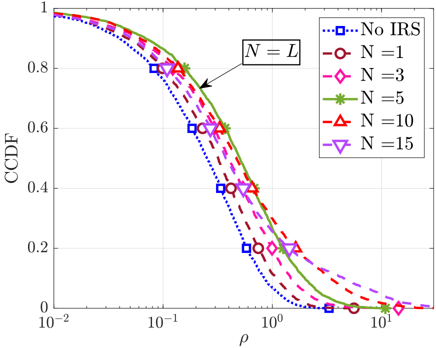


Directional response of IRS

IRS Can Only Help the OOB Operator: Numerical Illustrations



High SNR



Channel gain, $L = 5$

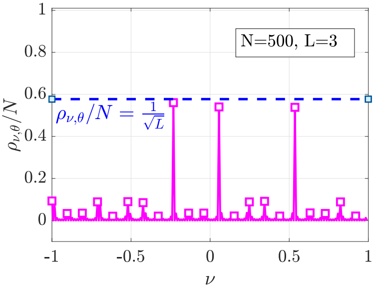
- IRS helps OOB UEs: increases the likelihood of virtual LoS
- OOB SNR: Linear scaling in N: $L \geq N$; else sub-linear in N
- OOB channel: with IRS stochastically dominates without IRS

Case 2: (L+)NLoS Scenarios - Directional Response of an IRS

- IRS is optimized to LoS and NLoS paths @ in-band UE
- Overall channel @ in-band UE:

$$h_k = h_{d,k} + \frac{N}{\sqrt{L}} \sum_{l=1}^{L} \gamma_{l,X}^{(1)} \gamma_{l,k}^{(2)} \dot{\mathbf{a}}_N^H(\omega_{X,k}^l)\boldsymbol{\theta}$$

- IRS focuses on multiple directions!



Spatial Amplitude Response of IRS: $\rho_{\nu,\theta} \triangleq N\mathbb{E}\left[\left|\dot{\mathbf{a}}^H(\nu)\boldsymbol{\theta}^{\text{opt}}\right|\right]$

$$\rho_{\nu,\theta} = \begin{cases} \Omega\left(\frac{N}{\sqrt{L}}\right) + o(N), & \text{if } \nu \in \left\{\omega_{X,k}^1, \dots, \omega_{X,k}^L\right\}, \\ o(N), & \text{if } \nu \in \boldsymbol{\Phi} \setminus \left\{\omega_{X,k}^1, \dots, \omega_{X,k}^L\right\}. \end{cases}$$

OOB Performance in (L+)NLoS Scenarios

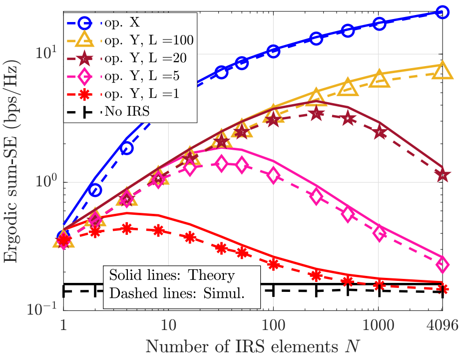
Acheivable Ergodic Sum-SE of OOB Operator

$$\bar{S}_L^{(Y)} \approx \begin{cases} \frac{1}{Q}\sum_{q=1}^{Q}\sum_{i=i_0}^{L}\frac{\binom{L}{i}\binom{N-L}{L-i}}{\binom{N}{L}} \\ \times \log_2\left(1+\left(\beta_{d,q}+i\frac{N^2}{L^2}\beta_{r,q}\right)\frac{P}{\sigma^2}\right), \text{if} \quad L<N, \\ \frac{1}{Q}\sum_{q=1}^{Q}\log_2\left(1+\left(\beta_{d,q}+N\beta_{r,q}\right)\frac{P}{\sigma^2}\right), \text{if} \quad L\geq N, \end{cases}$$

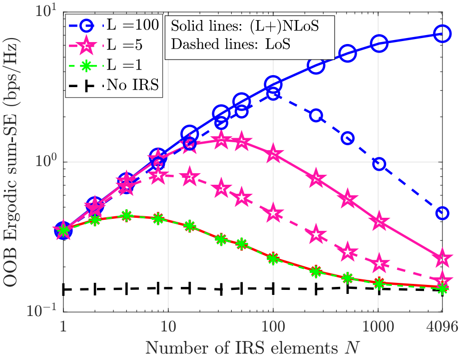
where $i_0 \triangleq \max\{0, 2L-N\}$.

- Linear scaling of SNR with N is possible when $L \geq \sqrt{N}$
- (L+)NLoS enhances OOB channels better than LoS

(L+)NLoS versus LoS Scenarios



(L+)NLoS Scenarios



Comparison

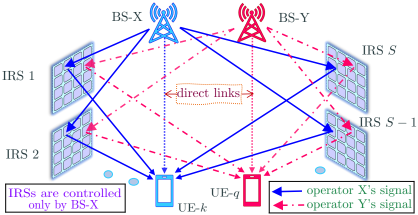
- Prob. of single path alignment via Stirling's Approx. $\approx \frac{L^2}{N} > \frac{L}{N}$
- Trade-off in (L+)NLoS scenarios: Better OOB performance **versus** higher signalling overheads for in-band operator

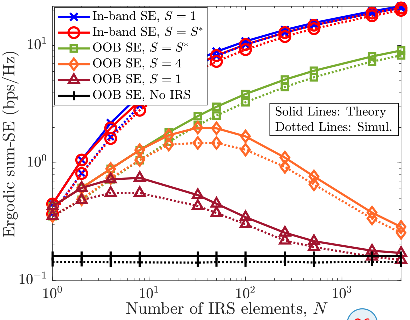
Distributed IRSs Offer Rich-Scattering Environments

- **Multiple IRSs help the OOB UEs better**: With S IRSs & M elements,

  Aligning prob. $: \frac{L}{N} \longrightarrow \frac{L}{M} = S\frac{L}{N}$

- OOB channel alignment follows a Binomial distribution
- Multiple IRSs can yield $\mathcal{O}(\log(N))$ scaling in the OOB SE almost surely
- Mimics a rich-scattering environment!
  - How many IRSs do we need?
- Distributed IRSs for spatial diversity!

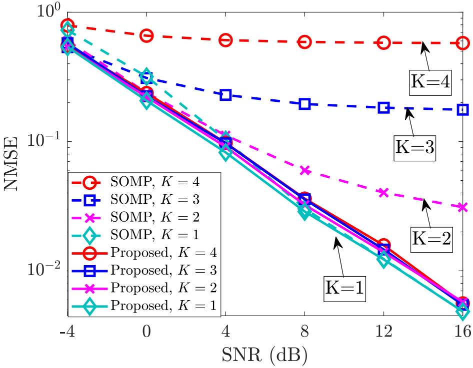




Low Complexity Channel Estimation for Distributed IRSs

Distributed IRSs offer benefits, yet at larger CSI estim. overheads!

- More channel coefficients implies larger pilot overheads!
- **Our Idea**: Leverage the subspace properties of the channel
- **Solution**: Joint *ESPRIT - MUSIC* based solution



- Our approach **does not scale with the number of IRSs**
- Requirement: # paths + 1 ≤ # BS antennas × # UE antennas
- CSI estim. with fewer pilots: Low time complexity

PART-III: IRS-Aided Wideband Beamforming

Research Question

- Wideband signals propagate in a frequency-selective channel
- IRS phase-shifters exhibit frequency-flat behaviour

Problem for PART - 3

How to **enable the IRS to beamform a wideband signal?**

Wideband beamforming

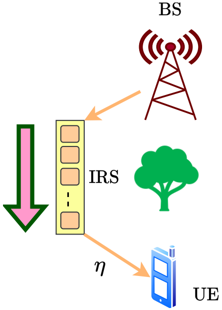
mmWave: Handling beam-split effects

Sub-6 GHz: Joint optimization across sub-bands

Revisiting the Channel Model in mmWave Bands

- **Setup**: An N-element IRS as a uniform linear array
- Channel impulse response from BS to nth element:

$$h_{1,n}(t) = \alpha\delta\left(t - \eta^{(1)} - (n-1)\frac{d}{c}\sin(\psi)\right) \times e^{-j2\pi f_c (n-1)\frac{d}{c}\sin(\psi)}$$



- Channel impulse response from nth element to UE:

$$h_{2,n}(t) = \gamma\delta\left(t - \eta^{(2)} + (n-1)\frac{d}{c}\sin(\omega)\right) \times e^{j2\pi f_c (n-1)\frac{d}{c}\sin(\omega)}$$

- Overall channel:

$$h(t) = \sum_{n=1}^{N} \theta_n \, h_{1,n}(t) \circledast h_{2,n}(t) \qquad (\circledast\text{—Linear convolution})$$

Large IRS Violates Narrowband Condition

- Simplified overall channel:

$$h(t) = \sum_{n=1}^{N} \alpha\gamma\delta\Big(t - \eta - \underbrace{(n-1)\frac{d}{c}\left(\sin(\psi) - \sin(\omega)\right)}_{\triangleq \tau_n}\Big) e^{-2\pi f_c (n-1)\frac{d}{c}\sin(\phi)},$$

  - η is total propagation delay - compensated by a timing offset
  - ϕ is the cascaded angle: $\phi \triangleq \sin^{-1}\left(\sin(\psi) - \sin(\omega)\right)$

- **Spatial delay spread**: $\Delta\tau = (N-1)\frac{d}{c}\left(\sin(\psi) - \sin(\omega)\right)$

- **Narrowband condition**: $\Delta\tau \ll T_s = 1/W$, so $\delta(t - \tau_n) \approx \delta(t)$

- **E.g.**: $N = 1024$, $W = 400$ MHz, $f_c = 30$ GHz, $\sin(\psi) - \sin(\omega) = 1$. Then, $\Delta\tau \approx 1.7 \times 10^{-8}$, $T_s = 2.5 \times 10^{-9}$: **Narrowband cond. fails**

- **Large arrays cause spatial-wideband effect**!

The Beam-Split: Curse of Spatial-Wideband Effect

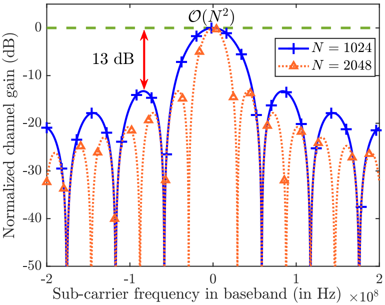
- Frequency response:

$$H(f) = \beta \sum_{n=1}^{N} \theta_n e^{-j\pi(n-1)\sin(\phi)\left(1+\frac{f}{f_c}\right)} = \sqrt{N}\beta\boldsymbol{\theta}^H \mathbf{a}_N\left(\sin^{-1}\left\{\left(1+\frac{f}{f_c}\right)\sin(\phi)\right\}\right)$$

- When the **IRS is tuned to** $f = 0$:
  - Channel gain on SC-k:

$$|H[k]|^2 = |\beta|^2 N^2 \operatorname{sinc}^2\left(N\frac{f_k}{2f_c}\sin(\phi)\right)$$

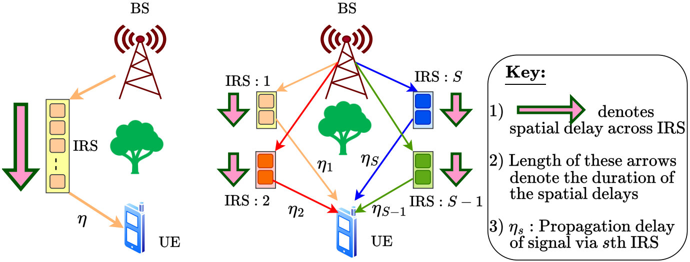
  - For prev. e.g., $N \leq 128$ for HPBW



- Array gain degrades on $f \neq 0$ SCs
  - **Beam-split effect** across frequencies
  - Limits the BW (or) # elements
  - **Problem: Handle beam-split effect!**

Approach I: Mitigating Beam-Split Effects

Idea: Multiple IRSs parallelize the spatial delays across IRSs



(a) Centralized IRS (b) Distributed IRSs

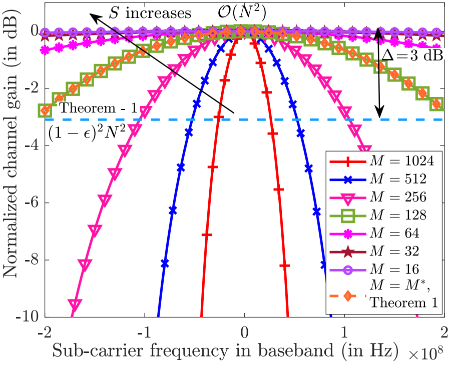
- Consider S IRSs with M elements each such that $SM = N$
- Modified delay spread: $\Delta\tau' \approx \frac{1}{S} \max_{s \in [S]} \left\{ (N-1)\frac{d}{c}\left(\sin(\psi_s) - \sin(\omega_s)\right) \right\}$
- Controlling M and S mitigates the B-SP effects to tolerable levels

How many elements per IRS?

ϵ-within beam-squint

We need a minimum gain of

$$\min_{k\in[K]} |H[k]|^2 \geq (1-\epsilon)^2 N^2$$



Maximum number of elements per IRS

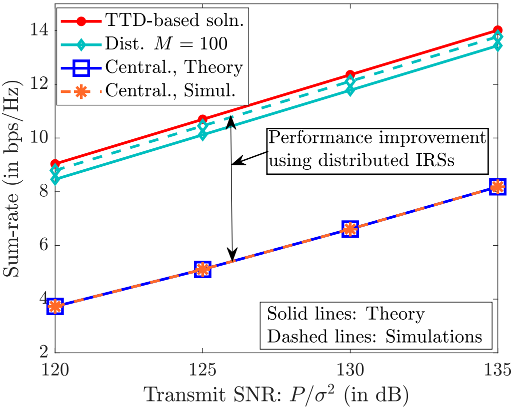
$$M \leq M^* \triangleq \min\left\{\max\left\{\left\lfloor \frac{4\sqrt{6\epsilon}}{\pi}\frac{f_c}{W}\right\rfloor, 1\right\}, N\right\}$$

Benchmarking with Existing Methods

- **Existing method**: True-time delay (TTD) units:

$$\theta_n(t) = e^{j\phi_n}\delta(t - \tau_n^{(l)})$$

- Completely cancels the channel delays
- **Demerits**:
  1. Hardware complexity
  2. More power due to high resolution TTD units
  3. Full-duplex operation



- Our solution performs equally well, but without TTDs
- Summary: a distributed IRS mitigates B-SP effects, achieves full array gain, and has low complexity

What Happens when Temporal Delay Spread $\neq 0$

Sum-rate: Distributed IRSs when TDS $\sim \mathcal{U}[0, T_0]$

$$\bar{R}_{\mathrm{D}} \geq \mathrm{R}_{\min} \triangleq \frac{1}{K + N_{\mathrm{CP}}^{\mathrm{D}}} \sum_{k=1}^{K} \log_2 \left(1 + \frac{p_k \sigma_{\mathrm{h}}^2}{\sigma^2} M^2 (1-\epsilon)^2 \times \left[S^2 \operatorname{sinc}^2(f_k T_0) + S\left(1 - \operatorname{sinc}^2(f_k T_0)\right)\right]\right)$$

- Gain variation: $\mathcal{O}(SM^2) \leq |H[k]|^2 \leq \mathcal{O}(S^2M^2 = N^2)$
- Distributed IRSs mitigate B-SP effect even if TDS $\neq 0$

Other Contributions

1. Positioning of IRSs to minimize the TDS
2. Angle diversity benefits to mitigate the B-SP effect
   - Dual benefits to mitigate the spatial-wideband/beam-split

Approach II: Exploiting Beam-Split Effects

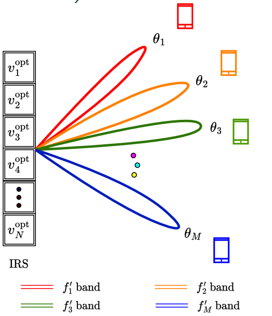
Conservation of energy: IRS focuses at different angles over the BW

- Opportunistic OFDMA over Q UEs:

$$\max_{\mathrm{SCH}(k,t)} \frac{1}{T}\sum_{t=1}^{T}\sum_{k=1}^{K}\frac{W}{K}\log_2\left(1+\frac{P}{K\sigma^2}|H(\mathrm{SCH}(k,t),t,f_k)|^2\right)$$

- **Solution**: Random IRS configuration + Max-rate scheduler
- **Premise:** On every SC, at least one UE will be near-optimal: multi-user diversity
- Random IRS configuration:

$$\phi_m(t) = 2\pi(m-1)a(t),\ a(t) \overset{\text{i.i.d.}}{\sim} \mathcal{U}[-1,1],\ t \in [T].$$

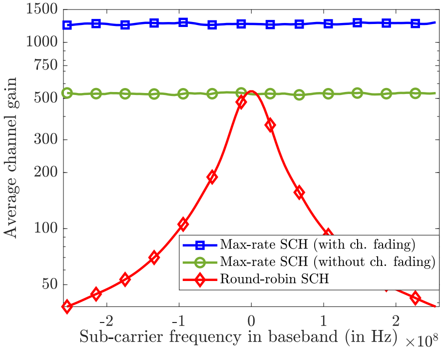


How many Users Are Sufficient?

- $(1-\epsilon)$-success event at UE-q on SC-k: $\mathcal{A}_{q,k}^{\epsilon} \triangleq \left\{ |H_q[k]|^2 \geq (1-\epsilon)N^2 \right\}$

Probability of success event, with K SCs and Q UEs

$$P_{\text{succ}}^{\epsilon} \triangleq \Pr\left(\bigcap_{k=1}^{K} \bigcup_{q=1}^{Q} \mathcal{A}_{q,k}^{\epsilon} \right) \geq 1 - K \left(1 - \frac{\sqrt{3\epsilon}}{\pi N \left(1 + \frac{W}{2f_c}\right)} \right)^{Q}$$



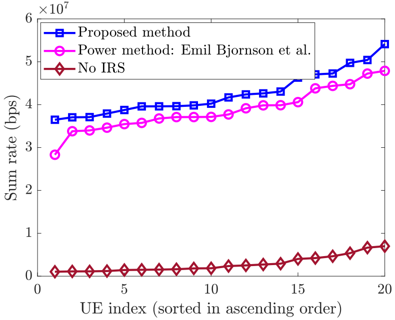
- Sufficient number of UEs:

$$Q \geq Q_{\min} \triangleq -\frac{\ln(K/\delta)}{\ln\left(1 - \frac{\sqrt{3\epsilon}}{\pi N(1+W/2f_c)}\right)}$$

- $Q_{\min}$ scales as $\mathcal{O}\left(\frac{NW}{2f_c}\right)$
- Array gain: $\mathcal{O}(N^2)$ over BW; multi-user diversity: $\mathcal{O}(\ln(Q))$

Wideband beamforming in Sub-6 GHz Bands

- Frequency-selectivity arises due to rich multi-path scattering



Joint Optimization across Sub-bands with Binary Phase Shift Constraints

$$\max_{\tilde{\boldsymbol{\theta}}} \quad R = \sum_{k=0}^{K-1} \log_2\left(1 + \left|\tilde{\boldsymbol{\theta}}^H \mathbf{h}_k\right|^2\right),$$

$$\text{s.t.} \quad \tilde{\boldsymbol{\theta}} \in \{-1,+1\}^{N+1}, \; \tilde{\boldsymbol{\theta}}[1] = 1.$$

- Proposed a low-pilot CSI estimator exploiting sparsity
- Optimization via the **majorization-minimization** technique
- Developed a series of equivalent optimization problems using matrix inequalities with closed-form solutions
- Our approach outperforms the existing methods

Summary

Take Home Message

1. **Optimal benefits without optimization**
   - Random phase sampling with opportunistic scheduling for optimal benefits
   - Moderate # UEs are sufficient: multi-user diversity
2. **IRS does not degrade the OOB performance**
   - IRS increases the richness of scattering
   - Multiple IRSs enable spatial diversity
   - Autonomy of all MOs in mmWave
3. **Large IRS breaks the narrowband condition**
   - Multiple IRSs for narrowband condition: mitigate beam-split
   - Beam-split enables multi-directional beamforming
   - Subspace methods for low-complexity CSI overheads

Publications

Journal Publications

- **L. Yashvanth**, and Chandra R. Murthy, **Performance Analysis of Intelligent Reflecting Surface Assisted Opportunistic Communications**, *IEEE Transactions on Signal Processing*, vol. 71, pp. 2056-2070, Mar. 2023.
- **L. Yashvanth**, and Chandra R. Murthy, **On the Impact of an IRS on the Out-of-Band Performance in Sub-6 GHz and mmWave Frequencies**, *IEEE Transactions on Communications*, vol. 72, no. 12, pp. 7417-7434, Dec. 2024.
- **L. Yashvanth**, and Chandra R. Murthy, **Distributed IRSs Always Benefit Every Mobile Operator**, *IEEE Wireless Communications Letters*, vol. 13, no. 11, pp. 2975-2979, Nov. 2024.
- **L. Yashvanth**, Chandra R. Murthy, and Bhaskar D. Rao, **Mitigating Spatial-Wideband and Beam-Split Effects Via Distributed IRSs: Design and Analysis**, under revision, *IEEE Transactions on Signal Processing*, Feb. 2025
- **L. Yashvanth**, Chandra R. Murthy, and Bhaskar D. Rao, **Spatial Correlation Aware Opportunistic Beamforming in IRS-aided Multi-User Systems**, under revision, *IEEE Wireless Communications Letters*, Mar. 2025
- S. Ghosh, **L. Yashvanth**, and Chandra R. Murthy, **Performance Analysis of Multi-IRS Aided Multiple Operator Systems at mmWave Frequencies**, revised and submitted to *IEEE Transactions on Communications*, May 2025

Conference Publications

- **L. Yashvanth**, Chandra R. Murthy and Deepak Battu, **Binary Intelligent Reflecting Surfaces Assisted OFDM Systems**, in *Proc. IEEE International Conference on Signal Processing and Communications (SPCOM)*, July 2022, Bangalore, India. pp. 1-5.
- **L. Yashvanth**, and Chandra R. Murthy, **Cascaded Channel Estimation for Distributed IRS Aided mmWave Massive MIMO Systems**, in *Proc. IEEE Global Communications Conference (GLOBECOM)*, Rio de Janeiro, Brazil, December 2022, pp. 717-723.
- **L. Yashvanth**, and Chandra R. Murthy, **Comparative Study of IRS Assisted Opportunistic Communications over I.I.D. and LoS Channels**, in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Rhodes Island, Greece, June 2023.
- **L. Yashvanth** and Chandra R. Murthy, **Does an IRS Degrade Out-of-Band Performance?**, in *Proc. IEEE 24th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, Shanghai, China, September 2023, pp. 216-220.
- **L. Yashvanth**, Chandra R. Murthy, and Bhaskar D. Rao, **Distributed IRSs Mitigate Spatial Wideband & Beam Split Effects**, in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Hyderabad, India, Apr. 2025.
- P. Siddhartha, **L. Yashvanth**, and Chandra R. Murthy, **Exploiting Beam-Split in IRS-aided Systems via OFDMA**, in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Hyderabad, India, Apr. 2025.

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SPC Lab

SPC LAB

Thank You... Questions?

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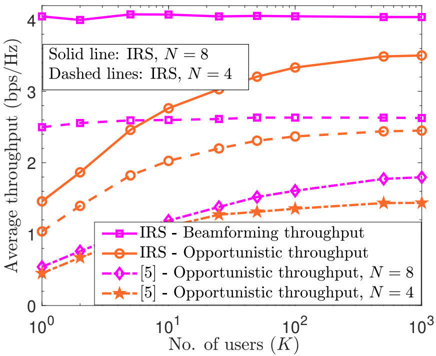
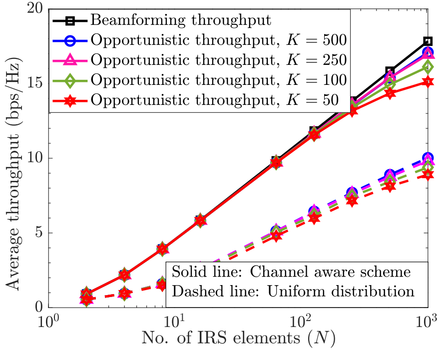
Backup Slides

Numerical Illustration



SE in i.i.d. channels



SE in LoS channels

- OC requires fewer UEs with LoS due to smaller DoR
- Choice of sampling distribution is important!

The Overall Protocol Exploiting the Spatial Correlation

Scheme 1: Spatial correlation-aware Opportunistic BF

Input: Correlation values: $\mathbf{U}$, $\mathbf{\Sigma}_\alpha$; BS-IRS link: $\mathbf{h}_1$.

for *time slot* $t = 1, 2, 3, \ldots$ **do**

/* Random sampling of IRS configurations */

Sample the random vector $\boldsymbol{\beta} \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma}_\alpha)$.

Set $\boldsymbol{\theta}_{\text{rand}} = \mathcal{F}\left(\left[1, (\mathbf{h}_1 \odot \mathbf{U}\boldsymbol{\beta})^T\right]^T\right)$, as per (4).

/* Towards identifying the *best* UE */

BS broadcasts a common pilot signal to every UE.

All UEs compute their PF metrics & feedback their identities to BS using *timer* schemes [11].

/* Proportional-fair scheduling of UEs */

Identity of the *best* UE-$k^*(t)$ arrives BS first, with

$$k^*(t) = \arg\max_{k=1,\ldots,K} R_k(t)/T_k(t).$$

The BS schedules UE-$k^*(t)$ for data transmission.

- Single pilot for CSI estim.
- **No** phase optimization
- **No** phase transportation

Success Rate of the Scheme: Proof Sketch

- **IRS vector**: $\theta = [e^{j\angle[\mathbf{f}']_1}, \ldots, e^{j\angle[\mathbf{f}']_N}]$,
- **Channel vector**: $\mathbf{h}_{f,k} = \sqrt{\beta_{r,k}}[[\mathbf{h}_{r,k}]_1, \ldots, [\mathbf{h}_{r,k}]_N]^T$

$$\Pr\left(\mathcal{E}_k^{\delta}\right) = \Pr\left(\left|\sum_{n=1}^{N} e^{-j\angle[\mathbf{f}']_n}[\mathbf{h}_{r,k}]_n\right|^2 \geq (1-\delta)\left|\sum_{n=1}^{N}\left|[\mathbf{h}_{r,k}]_n\right|\right|^2\right)$$

$$\overset{\text{can be shown}}{\geq} \Pr\left(\left|\sum_{m=1}^{M}\sqrt{\lambda_m}[\tilde{\mathbf{h}}_{2,k}]_m\right|^2 \geq (1-\delta)\left\|\mathbf{\Lambda}^{1/2}\tilde{\mathbf{h}}_{2,k}\right\|_1^2\right)$$

(Probability of reverse triangle inequality)

$$\overset{\|\mathbf{x}\|_1 \leq \sqrt{M}\|\mathbf{x}\|_2}{\geq} \Pr\left(\hat{\mathbf{h}}_k^H \mathbf{E}\hat{\mathbf{h}}_k \Big/ (\hat{\mathbf{h}}_k^H\hat{\mathbf{h}}_k) \geq (1-\delta)M\right)$$

(Tail probability of a Rayleigh quotient)

- Definitions: $\hat{\mathbf{h}}_k \triangleq \mathbf{\Lambda}^{1/2}\tilde{\mathbf{h}}_{2,k}$, $\mathbf{E} \triangleq \mathbf{1}_M\mathbf{1}_M^H$

Rayleigh Quotient of a Hetroscedastic Gaussian vector

On the Tail Probability of a Rayleigh Quotient

Let $\mathbf{A} \in \mathbb{C}^{L \times L}$ be a Hermitian rank-1 matrix, and $\alpha \in \mathbb{R}$ be such that $0 < \alpha < \|\mathbf{A}\|_F$. If $\mathbf{x} \in \mathbb{C}^L \sim \mathcal{CN}(\mathbf{0}, \mathbf{R})$ and $\mathbf{R}$ has full rank, the Rayleigh quotient of $\mathbf{A}$ w.r.t. $\mathbf{x}$ obeys

$$\Pr\left(\frac{\mathbf{x}^H \mathbf{A} \mathbf{x}}{\mathbf{x}^H \mathbf{x}} \geq \alpha\right) \geq \prod_{l=1}^{L} 1 \Big/ \left(1 + \frac{\alpha}{(\|\mathbf{A}\|_F - \alpha)} \cdot \frac{\lambda_{\mathbf{x},l}}{\lambda_{\mathbf{x},L}}\right),$$

where $\lambda_{\mathbf{x},1} \geq \ldots \geq \lambda_{\mathbf{x},L}$ are the ordered eigenvalues of $\mathbf{R}$.

Proof outline: The required probability is $\Pr\left(\mathbf{x}^H (\mathbf{A} - \alpha \mathbf{I}_L) \mathbf{x} \geq 0\right)$.

- Decoupling the quadratic form using EVD
- Total law of probability; orthogonal projection; Rayleigh-Ritz Thm.
- Independence of latent RVs, and MGF of exponential RVs

Rayleigh Quotient of Heteroscedastic Gaussian Vectors

- **How many users are sufficient?**

Pre-requisite:

On the Tail Probability of a Rayleigh Quotient

Let $\mathbf{A} \in \mathbb{C}^{L \times L}$ be a Hermitian rank-1 matrix, and $\alpha \in \mathbb{R}$ be such that $0 < \alpha < \|\mathbf{A}\|_F$. If $\mathbf{x} \in \mathbb{C}^L \sim \mathcal{CN}(\mathbf{0}, \mathbf{R})$ and $\mathbf{R}$ has full rank, the Rayleigh quotient of $\mathbf{A}$ w.r.t. $\mathbf{x}$ obeys

$$\Pr\left(\frac{\mathbf{x}^H \mathbf{A} \mathbf{x}}{\mathbf{x}^H \mathbf{x}} \geq \alpha\right) \geq \prod_{l=1}^{L} \frac{1}{\left(1 + \frac{\alpha}{(\|\mathbf{A}\|_F - \alpha)} \cdot \frac{\lambda_{\mathbf{x},l}}{\lambda_{\mathbf{x},L}}\right)},$$

where $\lambda_{\mathbf{x},1} \geq \ldots \geq \lambda_{\mathbf{x},L}$ are the ordered eigenvalues of $\mathbf{R}$.

Rayleigh Quotient of Heteroscedastic Gaussian Vectors: Proof

Proof: Let $P_\alpha \triangleq \Pr\left(\mathbf{x}^H\mathbf{A}\mathbf{x} \geq \alpha\mathbf{x}^H\mathbf{x}\right) = \Pr\left(\mathbf{x}^H\left(\mathbf{A} - \alpha\mathbf{I}_L\right)\mathbf{x} \geq 0\right)$. Define $\mathbf{B} \triangleq \mathbf{A} - \alpha\mathbf{I}_L$, with spectral decomposition: $\mathbf{B} = \mathbf{V}\mathbf{\Gamma}\mathbf{V}^H$. Then,

$$P_\alpha = \Pr\left(\mathbf{x}^H\mathbf{V}\mathbf{\Gamma}\mathbf{V}^H\mathbf{x} \geq 0\right) \overset{(a)}{=} \Pr\left(\tilde{\mathbf{x}}^H\mathbf{\Gamma}\tilde{\mathbf{x}} \geq 0\right) \overset{(b)}{=} \Pr\left(\sum_{l=1}^{L}\gamma_l\left|[\tilde{\mathbf{x}}]_l\right|^2 \geq 0\right).$$

Since $\mathbf{A}$ is Hermitian, $\mathbf{A} = \mathbf{a}\mathbf{a}^H$ for some $\mathbf{a} \in \mathbb{C}^L$. So, the eigenvalues of $\mathbf{B}$ are $\gamma_1 = \|\mathbf{a}\|_2^2 - \alpha > 0$, and $\gamma_2 = \gamma_3 \ldots = \gamma_L = -\alpha < 0$. Using this,

$$\begin{aligned} P_\alpha &= \Pr\left(\left|[\tilde{\mathbf{x}}]_1\right|^2 \geq \frac{\alpha}{\|\mathbf{a}\|_2^2 - \alpha}\sum_{l=2}^{L}\left|[\tilde{\mathbf{x}}]_l\right|^2\right) \\ &= \mathbb{E}_{[\tilde{\mathbf{x}}]_2,\ldots,[\tilde{\mathbf{x}}]_L}\left[\Pr\left(\left|[\tilde{\mathbf{x}}]_1\right|^2 \geq \frac{\alpha}{\|\mathbf{a}\|_2^2 - \alpha}\sum_{l=2}^{L}\left|[\tilde{\mathbf{x}}]_l\right|^2 \,\middle|\, [\tilde{\mathbf{x}}]_2, \ldots, [\tilde{\mathbf{x}}]_L\right)\right]. \end{aligned}$$

Proof Contd...

Let us decompose $\mathbf{x} = \mathbf{R}^{1/2}\mathbf{x}'$, where $\mathbf{x}' \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_L)$ and $\mathbf{R}^{1/2} = \mathbf{U}_{\mathbf{x}}\mathbf{\Lambda}_{\mathbf{x}}^{1/2}$ $\implies$ $\tilde{\mathbf{x}} = \mathbf{W}^H\mathbf{\Lambda}_{\mathbf{x}}^{1/2}\mathbf{x}'$, where $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_L]$ is a unitary matrix. As a result, $[\tilde{\mathbf{x}}]_1 \sim \mathcal{CN}(0, \sum_{l=1}^{L} \lambda_{\mathbf{x},l}|[\mathbf{w}_1]_l|^2)$.

Since $\|\mathbf{A}\|_F = \|\mathbf{a}\|_2^2$, and $\sum_{l=1}^{L} \lambda_{\mathbf{x},l}|[\mathbf{w}_1]_l|^2 \geq \lambda_{\mathbf{x},L}$, we lower bound P_α as

$$P_\alpha \geq \mathbb{E}_{[\tilde{\mathbf{x}}]_2,\ldots,[\tilde{\mathbf{x}}]_L}\left[e^{-\left(\alpha/\left((\|\mathbf{A}\|_F-\alpha)\lambda_{\mathbf{x},L}\right)\right)\sum\limits_{l=2}^{L}|[\tilde{\mathbf{x}}]_l|^2}\right].$$

Now, define $\tilde{\mathbf{x}}_{(1)} \triangleq [[\tilde{\mathbf{x}}]_2, \ldots, [\tilde{\mathbf{x}}]_L]^T$, $\mathbf{W}_{(1)} \triangleq [\mathbf{w}_2, \ldots, \mathbf{w}_L]$. Then,

$$\sum_{l=2}^{L} |[\tilde{\mathbf{x}}]_l|^2 = \left\|\tilde{\mathbf{x}}_{(1)}\right\|_2^2 = \mathbf{x}'^H\mathbf{\Lambda}_{\mathbf{x}}^{1/2}\mathbf{W}_{(1)}\mathbf{W}_{(1)}^H\mathbf{\Lambda}_{\mathbf{x}}^{1/2}\mathbf{x}' \overset{(c)}{\leq} \left\|\mathbf{\Lambda}_{\mathbf{x}}^{1/2}\mathbf{x}'\right\|_2^2,$$

where we noted that $\mathbf{W}_{(1)}\mathbf{W}_{(1)}^H$ is an orthogonal projector so that its eigenvalues are either 0 or 1; then used the Rayleigh-Ritz Theorem.

Proof Contd...

So, we further bound P_α as

$$
\begin{aligned}
P_\alpha &\geq \mathbb{E}_{[\mathbf{x}']_2,\ldots,[\mathbf{x}']_L}\left[e^{-\left(\alpha\big/\left((\|\mathbf{A}\|_F-\alpha)\lambda_{\mathbf{x},L}\right)\right)\sum\limits_{l=2}^{L}\lambda_{\mathbf{x},l}|[\mathbf{x}']_l|^2}\right] \\
&\overset{(d)}{=} \prod_{l=1}^{L}\mathbb{E}_{[\mathbf{x}']_l}\left[e^{-\left(\alpha\big/\left((\|\mathbf{A}\|_F-\alpha)\lambda_{\mathbf{x},L}\right)\right)\lambda_{\mathbf{x},l}|[\mathbf{x}']_l|^2}\right] \\
&\overset{(e)}{=} \prod_{l=1}^{L}\left(1\bigg/\left\{1+\frac{\alpha}{(\|\mathbf{A}\|_F-\alpha)}\cdot\frac{\lambda_{\mathbf{x},l}}{\lambda_{\mathbf{x},L}}\right\}\right),
\end{aligned}
$$

where in (d), we used the independence of $\{[\mathbf{x}']_l\}_{l=1}^{L}$; in (e), we used the expression for the moment generating function of the exponential random variables $|[\mathbf{x}']_l|^2, l \in [L]$. This completes the proof.

What is the Success Rate of the Scheme?

Success-event at UE-k: $\mathcal{E}_k^\delta \triangleq \left\{ |\boldsymbol{\theta}^H \mathbf{h}_{f,k}|^2 \geq (1-\delta) \|\mathbf{h}_{f,k}\|_1^2 \right\}, \delta \in (0,1)$

Success Probability of the Scheme using a PF Scheduler

$$P_{\mathrm{succ}} \geq 1 - \left(1 - \prod_{m=1}^{M} \frac{1}{1 + \frac{1-\delta}{\delta} \cdot \frac{\lambda_m}{\lambda_M}} \right)^K ,$$

where $\lambda_1 \geq \ldots \geq \lambda_M$ are the non-zero eigenvalues of $\boldsymbol{\Sigma}$.

Proof: The Success event of the scheme is

$$P_{\mathrm{succ}} = \Pr\left(\bigcup_{k=1}^{K} \mathcal{E}_k^\delta \right) \overset{(a)}{=} 1 - \prod_{k=1}^{K} \left(1 - \Pr\left(\mathcal{E}_k^\delta \right) \right) \qquad \textbf{(A)}$$

where (a) is due to the independence of channels across the K UEs.

Success Rate of the Scheme: Proof

- **IRS vector**: $\theta = [e^{j\angle[\mathbf{f}']_1}, \ldots, e^{j\angle[\mathbf{f}']_N}]$,
- **Channel vector**: $\mathbf{h}_{f,k} = \sqrt{\beta_{r,k}}[[\mathbf{h}_{r,k}]_1, \ldots, [\mathbf{h}_{r,k}]_N]^T$.

$$\begin{aligned}\Pr\left(\mathcal{E}_k^\delta\right) &= \Pr\left(\left|\sum_{n=1}^{N} e^{-j\angle[\mathbf{f}']_n}[\mathbf{h}_{r,k}]_n\right|^2 \geq (1-\delta)\left|\sum_{n=1}^{N} |[\mathbf{h}_{r,k}]_n|\right|^2\right)\\ &= \Pr\left(\left|\sum_{n=1}^{N} e^{-j\angle[\mathbf{f}]_n}[\mathbf{h}_{2,k}]_n\right|^2 \geq (1-\delta)\,\|\mathbf{h}_{2,k}\|_1^2\right),\end{aligned}$$

where we used the form of $\mathbf{f}'$ and $|[\mathbf{h}_1]_n| = 1$. Recall $\mathbf{h}_{2,k} = \mathbf{U}\boldsymbol{\Lambda}^{1/2}\tilde{\mathbf{h}}_{2,k}$, $\mathbf{f} = \mathbf{U}\boldsymbol{\Lambda}^{1/2}\tilde{\mathbf{f}}$. Since the channel distributions are invariant to multiplication by a unitary matrix, WLOG, we let $\mathbf{U} = [\mathbf{e}_1, \ldots, \mathbf{e}_M]$.

$$\Pr\left(\mathcal{E}_k^\delta\right) = \Pr\left(\left|\sum_{m=1}^{M} e^{-j\angle[\tilde{\mathbf{f}}]_m}\sqrt{\lambda_m}[\tilde{\mathbf{h}}_{2,k}]_m\right|^2 \geq (1-\delta)\left\|\mathbf{\Lambda}^{1/2}\tilde{\mathbf{h}}_{2,k}\right\|_1^2\right)$$

$$\geq \Pr\left(\left|\sum_{m=1}^{M}\sqrt{\lambda_m}[\tilde{\mathbf{h}}_{2,k}]_m\right|^2 \geq (1-\delta)M\left\|\mathbf{\Lambda}^{1/2}\tilde{\mathbf{h}}_{2,k}\right\|_2^2\right),$$

where we dropped $e^{-j\angle[\tilde{\mathbf{f}}]_m}$ because $\angle[\tilde{\mathbf{f}}]_m$ is uniformly distributed in $[0, 2\pi)$ and independent of $\angle[\tilde{\mathbf{h}}]_m$, and we used $\|\mathbf{x}\|_1 \leq \sqrt{M}\|\mathbf{x}\|_2$. So,

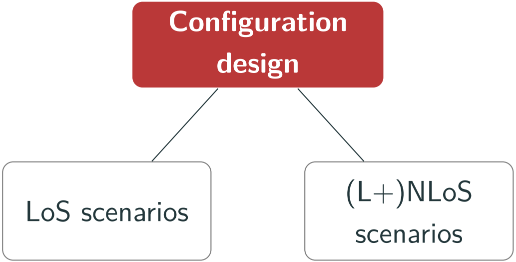
$$\Pr\left(\mathcal{E}_k^\delta\right) \geq \Pr\left(\hat{\mathbf{h}}_k^H\mathbf{E}\hat{\mathbf{h}}_k \Big/ (\hat{\mathbf{h}}_k^H\hat{\mathbf{h}}_k) \geq (1-\delta)M\right),$$

where $\hat{\mathbf{h}}_k \triangleq \mathbf{\Lambda}^{1/2}\tilde{\mathbf{h}}_{2,k}$, $\mathbf{E} \triangleq \mathbf{1}_M\mathbf{1}_M^H$ with $\|\mathbf{E}\|_F = M > (1-\delta)M > 0$. Then, using the tail probability of the Rayleigh quotient result,

$$\Pr\left(\mathcal{E}_k^\delta\right) \geq \prod_{m=1}^{M}\left(1 \Big/ \left\{1 + \frac{1-\delta}{\delta}\cdot\frac{\lambda_m}{\lambda_M}\right\}\right).$$

Using this in (**A**), we get the desired result.

Two Design Choices



1. **LoS scenario**:
   - IRS is aligned to the strongest in-band path - **virtual LoS**
   - IRS configuration is a phasor
   - Control overheads do not scale with N
2. **(L+)NLoS scenario**:
   - IRS is aligned to add to all the paths @ in-band UE coherently
   - No pattern in the IRS configuration vector
   - Control overheads scales as $\mathcal{O}(N)$

Ergodic Spectral Efficiency in LoS Scenarios

Acheivable Ergodic Sum-SE of the Operators

$$\textbf{MO-X:}\ \bar{S}_1^{(X)} \approx \frac{1}{K}\sum_{k=1}^{K}\log_2\left(1+\left[N^2\beta_{r,k}c_1+Nc_2+\beta_{d,k}\right]\frac{P}{\sigma^2}\right)$$

$$\textbf{MO-Y:}\ \bar{S}_1^{(Y)} \approx \frac{1}{Q}\sum_{q=1}^{Q}\left(\frac{\bar{L}}{N}\log_2\left(1+\left[\frac{N^2}{\bar{L}}\beta_{r,q}+\beta_{d,q}\right]\frac{P}{\sigma^2}\right)\right.$$
$$\left.+\left(1-\frac{\bar{L}}{N}\right)\log_2\left(1+\beta_{d,q}\frac{P}{\sigma^2}\right)\right),$$

where $\bar{L}=\min\{L,N\}$, and $L \triangleq L_1L_2$.

- IRS helps OOB UEs: increases the likelihood of virtual LoS
- IRS does not degrade the OOB performance

How Does an IRS Alter the Instantaneous Performance?

Outage Probability at OOB UE-q: $\Pr(|h_q|^2 < \rho)$

$$1 - e^{-\frac{\rho}{\beta_{d,q}}} - \frac{\bar{L}}{N}\left(\frac{\bar{L}e^{\frac{\bar{L}\beta_{d,q}}{N^2\beta_{r,q}}}}{N^2\beta_{r,q}}\mathcal{I}_0\left(\rho; \beta_{d,q}, \frac{N^2}{\bar{L}}\beta_{r,q}\right) - e^{-\frac{\rho}{\beta_{d,q}}}\right),$$

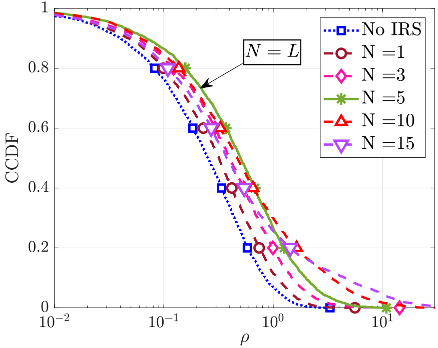
where $\bar{L} = \min\{L, N\}$, $\mathcal{I}_0(x; c_1, c_2) \triangleq \int_{c_1}^{\infty} e^{-\left(\frac{x}{t}+\frac{t}{c_2}\right)}\mathrm{d}t.$

- **OOB channel gain indicator**: $G_1 = |h_q|^2$ in the presence of IRS, $G_0 = |h_q|^2$ in the absence of IRS:

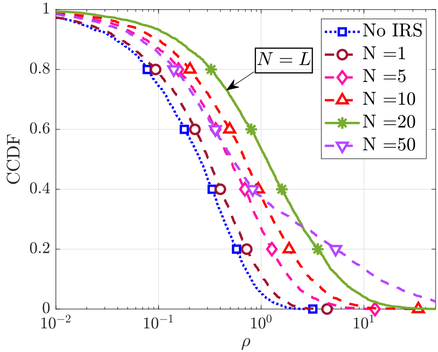
$$\Pr(G_1 > \rho) \geq \Pr(G_0 > \rho), \quad \forall \rho \in \mathbb{R}^+$$

- The OOB channel in the **presence of IRS stochastically dominates** the OOB channel in the **absence of IRS**

Numerical Illustration



L=5

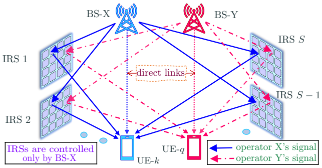


L=20

- Achievable **OOB channel gain, SE, outage probability** are **better in the presence of an IRS than in its absence**

Improving OOB Performance via Distributed IRS in mmWave

- Can we improve the OOB performance?
  - Deploy multiple smaller and distributed IRSs
  - # paths increases, resolution of the array decreases



1. **Key Motivation**:
   - With multiple IRSs, the chances that the IRS benefits an OOB UE increase

2. **Dual Benefits**:
   - Deploy S IRSs each with M elements: $N = SM$
   - Probability of alignment @ IRS modifies as $\frac{L}{N} \longrightarrow \frac{L}{M} = S\frac{L}{N}$
   - Multiple IRSs can align independently at an OOB UE

Ergodic Spectral Efficiency with Distributed IRSs

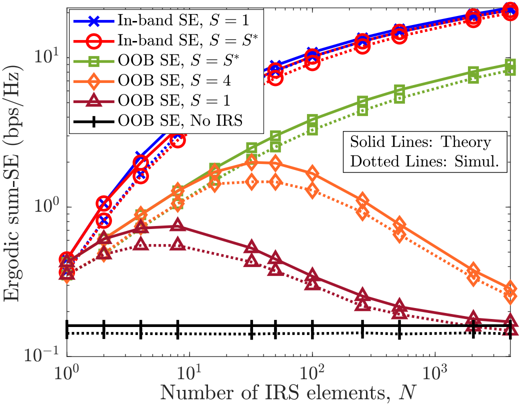
Acheivable Ergodic Sum-Spectral Efficiency of the Operators

$$\bar{S}_M^{(X)} \approx \frac{1}{K}\sum_{k=1}^{K}\log_2\left(1+\left[N^2 c_1 + N c_2 + \beta_{d,k}\right]\frac{P}{\sigma^2}\right),$$

$$\bar{S}_M^{(Y)} \approx \begin{cases} \frac{1}{Q}\sum_{q=1}^{Q}\sum_{\mathfrak{s}=0}^{S}\frac{S!}{(S-\mathfrak{s})!\mathfrak{s}!}\left(\frac{L}{M}\right)^{\mathfrak{s}}\left(1-\frac{L}{M}\right)^{S-\mathfrak{s}} & \\ \quad\times\log_2\left(1+\left[\frac{\mathfrak{s}M^2}{L}\beta_{r,q}+\beta_{d,q}\right]\frac{P}{\sigma^2}\right), & \text{if } L < M, \\ \frac{1}{Q}\sum_{q=1}^{Q}\log_2\left(1+\left(\beta_{d,q}+N\beta_{r,q}\right)\frac{P}{\sigma^2}\right), & \text{if } L \geq M \end{cases}$$

- OOB Benefits are governed by a Binomial distribution!
- How to obtain $\mathcal{O}(\log(N))$ scaling of OOB SNR almost surely?

- Deploy sufficiently many IRSs!



How many IRSs?

Let $\delta^* \triangleq \min\{1, \log_N L\}$.

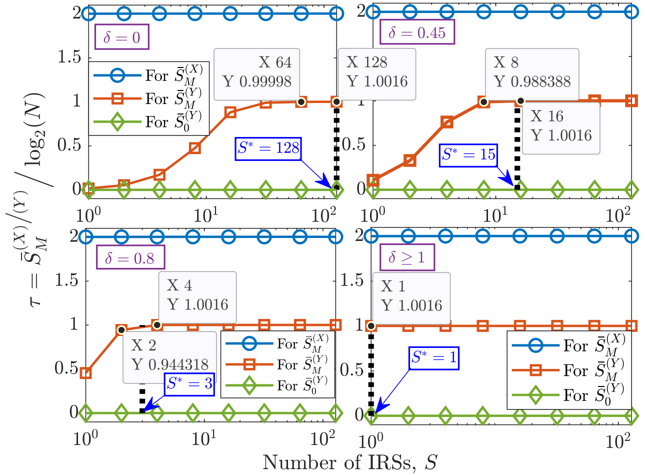
Then

$$S \geq S^* \triangleq \lceil N^{1-\delta^*} \rceil$$

- With sufficiently many IRSs
  - Obtain full-scattering gain of $\mathcal{O}(\log(N))$ like sub-6 GHz, a.s.
- **Distributed IRSs offer rich-scattering in mmWave**!

How Much Rich-Scattering Can be Obtained?

- We can show that OOB SE scales as $\mathcal{O}(\tau \log_2(N))$, $\tau \in [0, 1]$
- $\delta = \log_N L$ indicates the "*richness of scattering*"



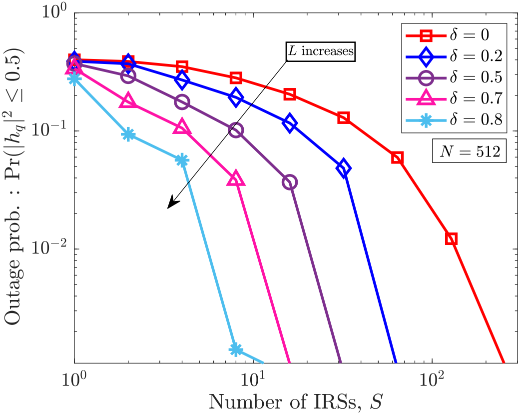
Spatial Diversity Benefits of Distributed IRSs

- What is the **outage prob. at a OOB UE**?

ρ-Outage Prob. with S IRSs

$$P^{\rho}_{out,q} = \left(1 - e^{-\rho/\beta_{d,q}}\right) \mathrm{P}_{\mathbf{0}}^{\mathbf{S}},$$

where P_0 is the OOB outage prob. obtained via 1 IRS



- Outage prob. reduces exponentially in the number of IRSs,
- **Distributed IRSs promote spatial diversity** at a OOB UE
- **Multiple IRSs help OOB MOs better than a single IRS** in both average and instantaneous senses

Low Complexity Channel Estimation for Distributed IRSs

- Distributed IRSs provide benefits, yet at larger CSI estim. overheads
- More channel coefficients implies larger pilot overheads!
- **Our Idea**: Leverage the subspace properties of the channel
- **Solution**: Joint *ESPRIT - MUSIC* based solution
- Shift in the overheads: # pilots $\longrightarrow$ # BS antennas

Algorithm 1: DoA estimation at the BS using ESPRIT

Input: $\widehat{\mathbf{R}}, M, N, K$

1. Perform the eigenvalue decomposition of $\widehat{\mathbf{R}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$ and obtain $(\mathbf{\Lambda}, \mathbf{U})$.
2. Select the K dominant eigen vectors of $\mathbf{U}$ to form a basis for signal sub-space, $\mathbf{U}_S$.
3. Define selection matrices $\mathbf{W}_1 \triangleq [\mathbf{I}_{MN-1} \mid \mathbf{0}_{MN-1}]$ and $\mathbf{W}_2 \triangleq [\mathbf{0}_{MN-1} \mid \mathbf{I}_{MN-1}]$.
4. Compute $\mathbf{U}_{S,1} = \mathbf{W}_1\mathbf{U}_S$ and $\mathbf{U}_{S,2} = \mathbf{W}_2\mathbf{U}_S$.
5. Obtain $\mathbf{\Omega} = \mathbf{U}^{\dagger}_{\mathbf{S},1}\mathbf{U}_{\mathbf{S},2}$.
6. Compute the eigen values of $\mathbf{\Omega}$ as $\xi_1, \xi_2, \xi_3, \ldots, \xi_K$.
7. Compute $\widehat{\psi}_k = \sin^{-1}\left(-\frac{\arg(\xi_k)}{\pi}\right)$, where $\arg(x) = \tan^{-1}\left(\frac{\mathcal{I}(x)}{\mathcal{R}(x)}\right)$, $k = 1, \ldots, K$

Output: $\left\{\widehat{\psi}_k\right\}_{k=1}^{K}$

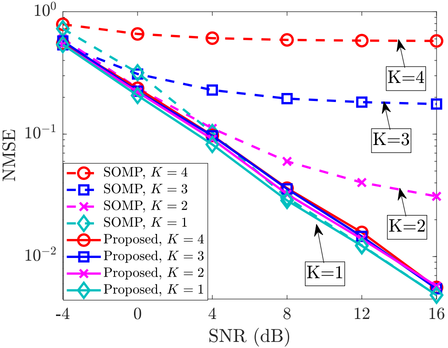
Algorithm 2: Estimation of the DoD from the UE using MUSIC

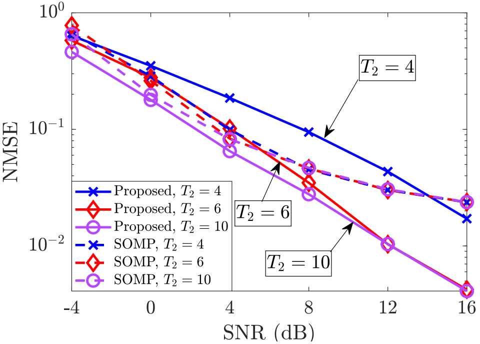
Input: $M, N, K, \{N_k\}_{k=1}^{K}, \left\{\widehat{\psi}_k\right\}_{k=1}^{K}, (\mathbf{\Lambda}, \mathbf{U}), d$

1. Select the $MN - \sum_k N_k$ eigen vectors of $\mathbf{U}$ corresponding to the least $MN - \sum_k N_k$ eigenvalues to form a basis for noise sub-space, $\mathbf{U}_N$.
2. **for** $k = 1$ *to* K **do**
3.   **for** $\phi = -\frac{\pi}{2}$ *to* $\frac{\pi}{2}$ *in steps of* d **do**
4. $\mathbf{P}_{\text{spec}}(k, \phi) = \frac{1}{\left\|\left\{[\mathbf{a}_N(\phi)\otimes\mathbf{I}_M]\mathbf{a}_M(\widehat{\psi}_k)\right\}^H\mathbf{U}_N\right\|_2^2}$.
5. $\left\{\widehat{\phi}_{i,k}\right\}_{i=1}^{N_k} = \underset{\phi}{\operatorname{argmax}}^{(N_k)}\ \mathbf{P}_{\text{spec}}(k, \phi)$.

Output: $\left\{\widehat{\phi}_{i,k}\right\}_{i=1,k=1}^{N_k,K}$.

Pilots Does Not Scale with # IRSs, # IRS elements





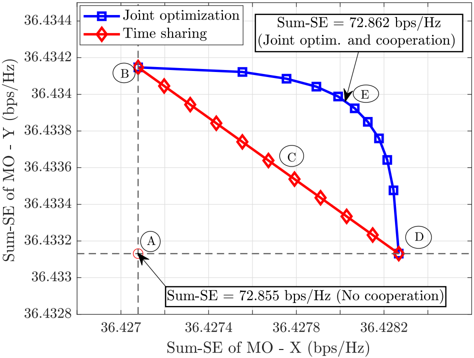
NMSE vs. # IRSs **NMSE vs. # Pilots**

- Our approach is **robust to number of IRSs**
- Need: # paths + 1 $\leq$ # BS ant. $\times$ # UE ant.
- CSI estim. with fewer pilots: Low time complexity

How much does Inter-MO Cooperation Help?

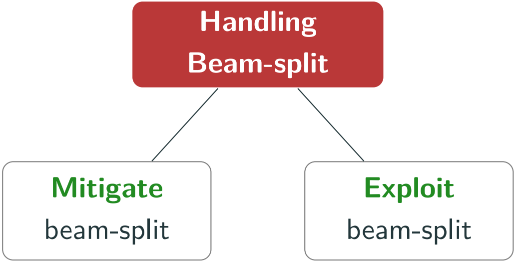
What happens when every MO deploys and controls an IRS?

- Each UE receives signals via the in-band IRS and OOB IRSs
- 2-MO system has 4 events
- 3 transmission schemes:
  1. MO-cooperation: joint optim.
  2. MO-cooperation: time-sharing
  3. No MO-cooperation
- **How helpful is cooperation?**



- Joint optimization performs the best!
- Cooperation helps very little due to the sparsity in the channel
- IRS does not degrade the OOB performance! Every MO can control its IRS autonomously

Proposed Approaches to Handle Beam-Split Effects



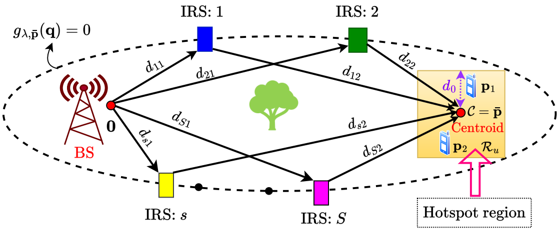
1. **Mitigating beam-split effects**:
   - Obtain full array gain @ a UE
   - **Idea**: Multiple IRSs reduce the spatial delay spread
2. **Exploiting beam-split effects**:
   - Obtain full array gain @ the network
   - **Idea**: Opportunistic OFDMA-based scheduling of multiple UEs

Where to Position the IRSs: Minimizing the TDS

- How to position the IRSs so that TDS is minimized?
- **Single UE**-case: Let $\mathbf{0}$, $\mathbf{p}$, $\{\mathbf{q}\}_{s=1}^{S}$ be BS, UE, IRS locations. We solve

$$\{\mathbf{q}_s^*\}_{s=1}^{S} = \arg \min_{\mathbf{q}_1,\ldots,\mathbf{q}_S} \left\{ \max_{1\le s\le S} \frac{(\|\mathbf{q}_s\|_2 + \|\mathbf{p}-\mathbf{q}_s\|_2)}{c} - \min_{1\le s\le S} \frac{(\|\mathbf{q}_s\|_2 + \|\mathbf{p}-\mathbf{q}_s\|_2)}{c} \right\}$$

- **Solution**: IRSs on an ellipse with BS and UE as foci
- **Multi UE**-case: Use an ellipse with BS & centroid of UE distribution as foci - The TDS scales linearly with the distance from the centroid



$$d_{11} + d_{12} = d_{21} + d_{22} = \ldots = d_{s1} + d_{s2} = \ldots = d_{S1} + d_{S2} = \lambda$$

Angle Diversity Benefits of Distributed IRSs

- Beam-split induced outage: $\mathbb{P}_\rho^k = \Pr\left(\{\phi_1, \dots, \phi_S\} : |H[k]|^2 \le \rho\right)$

Outage Probability with Distributed and Centralized IRS

D-IRS: $\mathbb{P}_\rho^k \le e^{-2S\left(\left\{1-\frac{\sqrt{\rho}}{SM\xi\sigma_{\mathrm{h}}^{\min}}\right\}\frac{24f_c^2}{\pi^2M^2f_k^2}-\mu_X^{\max}\right)^2}, \rho \in (0, \rho_{\mathrm{D}}^k),$

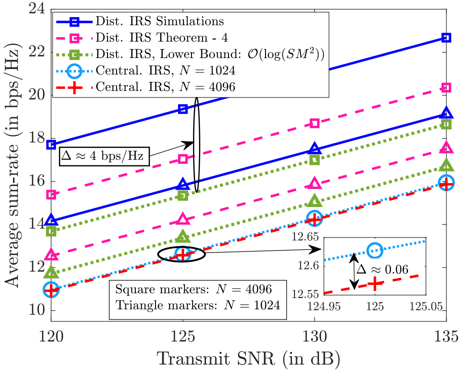
where $\rho_{\mathrm{D}}^k = N^2\xi\,(1-\delta)$, and $\mu_X^{\max}$ is max-mean angular spread

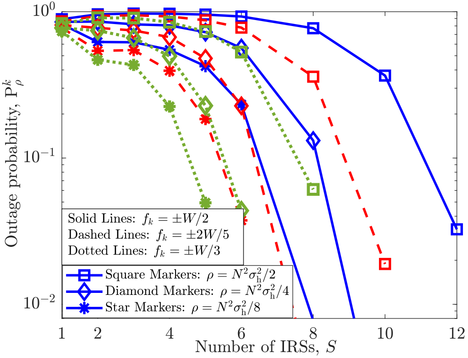
C-IRS: $\mathbb{P}_\rho^k \ge 1 - \frac{1}{\phi_0}\left\{\sin^{-1}\left(\frac{2f_c\sigma_{\mathrm{h}}}{\pi f_k\sqrt{\rho}}\right)\right\}, \rho \ge \frac{4f_c^2\sigma_{\mathrm{h}}^2}{\pi^2 f_k^2 c_0^2} \sim \mathcal{O}(N^2),$

where $c_0 = \sin(\phi_0)$, and $[-\phi_0, \phi_0]$ is the angular range

- Outage prob. exponentially decays in S - **Angle diversity**
- Outage prob. is bounded away from 0 for central. IRS

Dual Benefits of Dist. IRSs to Mitigate SW/B-SP Effects





Sum-rate vs. SNR

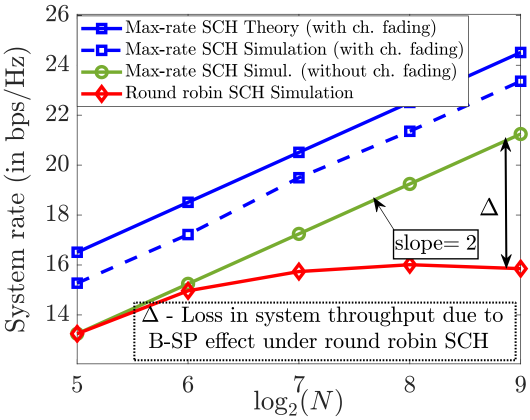
Outage Prob. vs. # IRSs

- **Summary**: Distributed IRSs mitigate B-SP effects in 2 ways:
  1. Parallelize the spatial delays and deploy IRSs on an ellipse
  2. Angle diversity due to independent paths

Achievable Throughput

Throughput of the Scheme

$$\lim_{Q\to\infty}\left(R^{(Q)} - \mathcal{O}\left\{W\log_2\left(1+\frac{\rho G_{\mathrm{Tx}}G_{\mathrm{Rx}}P}{K\sigma^2}N^2\ln Q\right)\right\}\right)=0$$



- Full array gain: $\mathcal{O}(N^2)$ over the full BW
- Multi-user diversity gain: $\mathcal{O}(\ln(Q))$

Directions of Future Study

- Opportunistic communications under user mobility: updating the sampling distribution
- Performance study of randomly configured IRSs under generalized schedulers?
- Channel estimation for IRS-aided multiple MO system
- Beam management using random beamforming